

Basic Principles of Medicine 1

Module : Musculoskeletal

DLA

Development of the limbs

Part 2 Clinical

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Recommended Reading

The Developing Human 11th edn – Clinically Oriented Embryology

Moore K.L, Persaud T.V.N, Torchia M.G.

Chapter 16 pages 341 - 54

Review early embryology DLA lecture slides

Watch this video:

<https://premiumbasicsscience.lovehealthlibrary.com/multimediaplayer.aspx?MultimediaID=19450537>

Pay special attention to the **green boxes** which contain the clinical correlates for this topic.

Special note

Some slides have a green background, this will not be covered in the video recording but are put so you have the information available

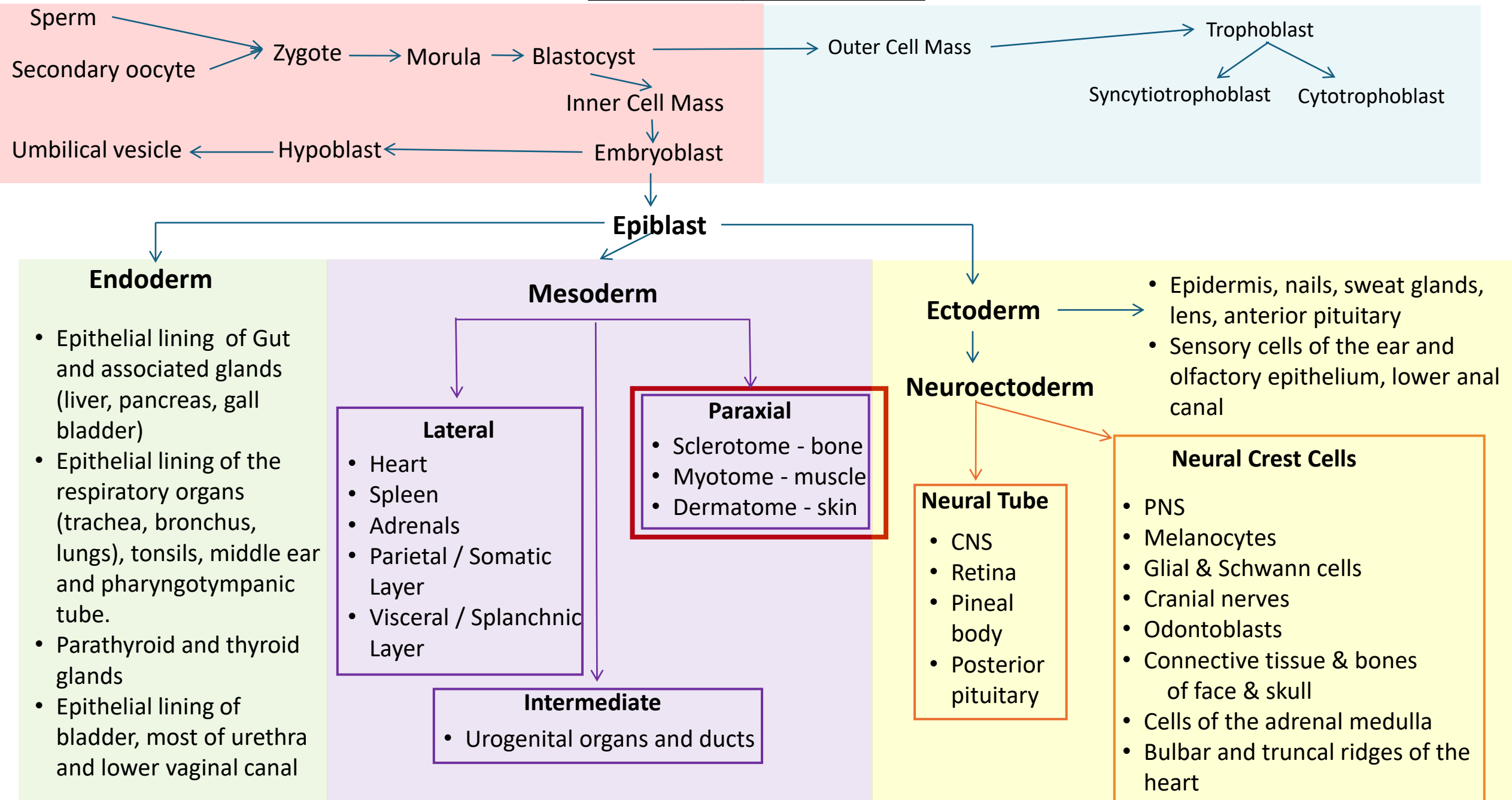
Important notes

Because of the orientation of the embryo, the terms dorsal and ventral do not have the same meaning as anterior and posterior.

Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1345	Explain the formation of the following: congenital talipes (clubfoot), amelia / meromelia, cutaneous syndactyly, osseous syndactyly, brachydactyly, polydactyly, ectrodactyly (cleft hand or partial adactyly) and limb deformities caused by amniotic bands.
SOM.MKII.BPM1.1.MSK.1.ANAT.1346	List and explain the clinical significance of the VACTERL association

The Big picture

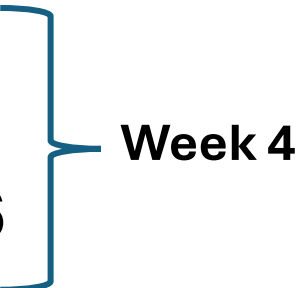


The journey of development

- Fertilization to gastrulation - Some primordial cell differentiation happens - **Weeks 1-3**

Organogenesis – Weeks 4 to 8

Everything happens synchronously with many simultaneous events

- Formation of the neural tube, spinal cord and nerves, somite differentiation (week 4 & 5)
 - Upper limb buds appear on day 24
 - Lower limb buds appear on day 25/26
 - Development of cartilage model for bone development appear in **week 5**
 - Ossification sites appear in **week 6**
 - Ossification of limb bones occur in **weeks 7 -12**
- 

Limb defects

- Four categories:
 - Reduction – part of the limb is missing
 - Duplication – supernumerary limb elements present
 - Dysplasia – fusion or disproportionate growth of limb elements
 - Deformation – damage due to some external force or something that affects the appearance of an otherwise normal limb

VACTERL Syndrome

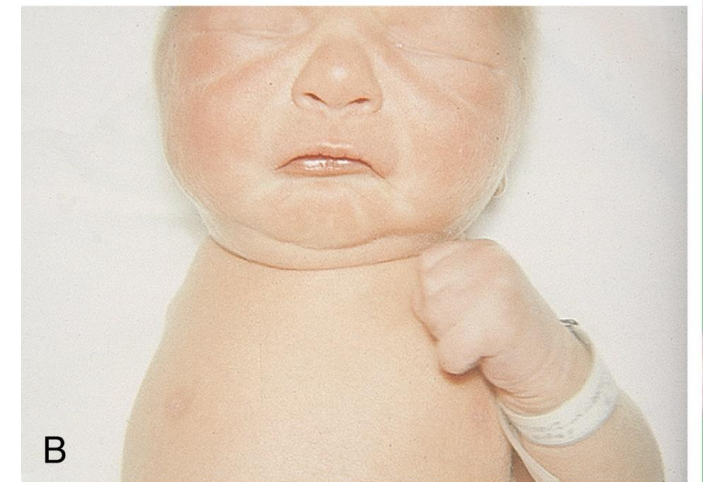
- Occurs in 1 in 10,000 to 40,000 births.
- Caused by the interaction of multiple genetic and environmental factors.
- VACTERL stands for:
 - Vertebral defects
 - Anal atresia
 - Cardiac defects
 - Tracheo-esophageal fistula
 - Renal anomalies
 - Limb abnormalities
- People diagnosed with VACTERL association typically have at least three of these characteristic features

Reduction: Amelia & Meromelia

- Amelia (B)
 - Complete absence of limbs
 - Suppression of limb bud development in the 4th week
- Meromelia (A)
 - Partial absence of limbs
 - Disturbance of limb development during 5th week

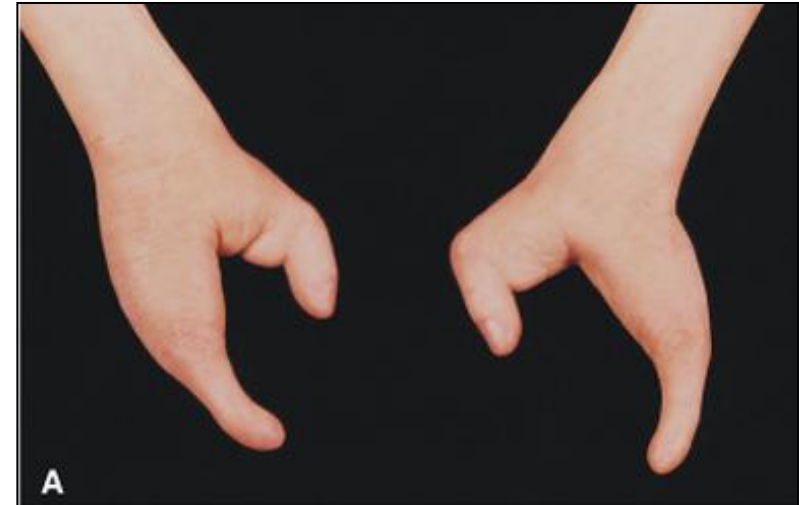
Defects caused by:

- Genetic factors
- Mutant genes
- Environmental factors
- Vascular disruption and ischemia



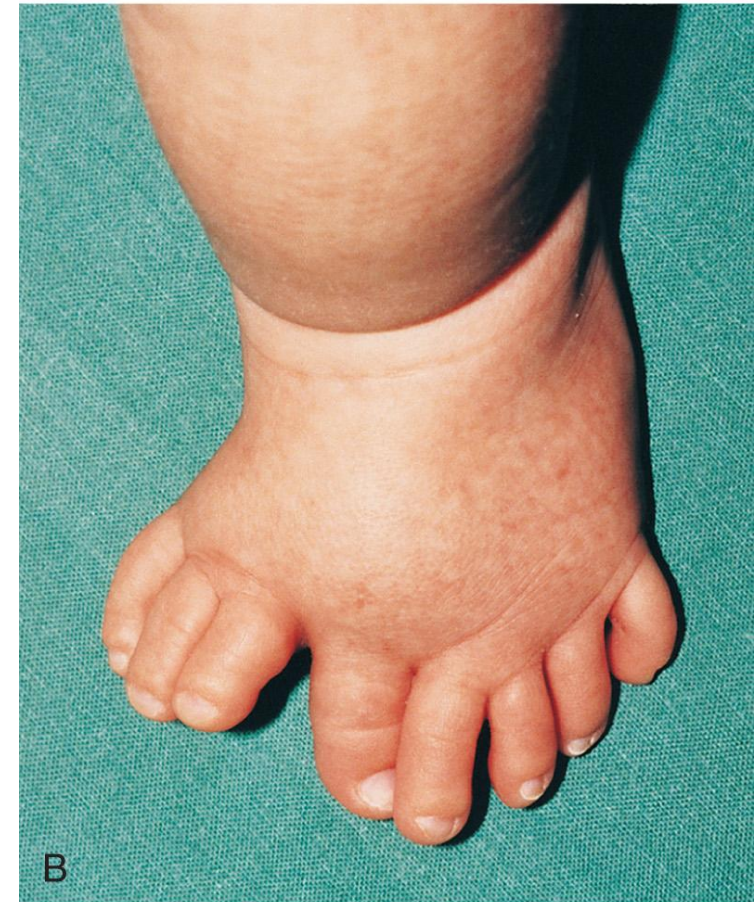
Reduction: Cleft Hand & Cleft Foot (Ectrodactyly)

- Absence of one or more central digits
 - Failure of development of one or more digital rays
- Remaining digits are partially or completely fused
- Rare anomaly
 - Lobster-claw deformities



Duplication: Polydactyly

- Supernumerary digits
- Disruption of the anteroposterior pattern
- Inherited as a dominant trait.



Dysplasia: Syndactyly

- 1:2200 births
- Cutaneous syndactyly (most common limb defect)
 - Webbing of the digits – failure of webs to degenerate between 2 or more digits
- Osseous syndactyly
 - Fusion of the bones – failure of development of notches between the digital rays



Deformation: Club Foot (Congenital Talipes)

- Talipes equinovarus (is most common type)
- Twice more frequently in males as in females.
- Sole of foot is turned medially and the foot is inverted.
- Multifactorial pattern of inheritance.



Deformation: Amniotic Band Syndrome

- Tears in the amnion result in amniotic bands
- May encircle fetal limbs & digits
 - Ring constrictions
 - Amputations
- Cause unknown
 - May be due to infection
 - ~1:1200 births

